Zhiyuan Xia

Research Interests: AI for Science, System Design & Modeling, Human-Computer Interaction
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Education Background

Tianjin University **Sep. 2024 - Present (Expected June 2027)**
Master of Engineering, Environmental Engineering

Focus: Machine Learning & Sustainable Development & Water Treatment
Awards: Merit-based Scholarship-Level A (Oct. 2024)

Georgia Institute of Technology **Aug. 2024 - Aug. 2025**
Master of Science, Environmental Engineering (Track: Data Analytics in Environment)

Courses: Environmental Modeling; Random Process; Data and Visual Analytics; Probabilistic Models;
GPA: 3.75/4.0

Zhejiang Sci-Tech University **Sep. 2016 - June 2020**
Bachelor of Engineering, Industrial Design (Excellent Engineering Program)

GPA: 85/100
Courses: Product Integration Design; Product Design Methods; Ergonomics
Awards: Industrial Design Contest of Zhejiang Province-First Prize (Oct. 2018)

Working Experience

Apple Greater China
Retail Team, Specialist **Sep. 2021 - Apr. 2022**
AI/ML Team, Siri Annotation Analyst **July 2020 - Aug. 2021**

- Validated and curated large-scale datasets for training production-level NLP models;
- Collaborated with ML engineers to optimize data annotation pipelines, improving data quality and team efficiency by 20%;

Research Experience

AI-Enabled Framework for Adaptive Nutrient Recovery in Urban Farming
Guided by Prof. Wenlong Zhang@GTSI & Prof. Peizhe Sun@TJU **Sep. 2025 - Present**

- **Proposal:** Addressing the "Form-Function Mismatch" that prevents cost-effective nutrient recovery from urban wastewater for hydroponic systems;
- **Method:** Co-authored a review paper that critically analyzes existing technologies and proposes a conceptual AI-enabled framework for adaptive, real-time process control;
- **Outcome:** Manuscript in preparation for submission to a high-impact journal.

AI for Science: Transformer-based Framework for Chemical Reaction Kinetics
Graduate Researcher, Tianjin University **Nov. 2025 - Present**

- **Proposal:** Predicting the non-linear reaction kinetics of single-atom catalysts from sparse and noisy experimental data;
- **Method:** Developed a **Transformer-based framework** (PyTorch) with K-Fold cross-validation and deployed it as an interactive Web App (Streamlit);
- **Outcome:** github.com/AdorableLake/SACs-Kinetics-Transformer

Multi-Agent Systems: Simulation of Auction-based Traffic Coordination
Project Developer, Independent Research **Dec. 2025 - Present**

- **Proposal:** Exploring decentralized coordination mechanisms for autonomous vehicles at unsignalized intersections to improve efficiency and fairness;
- **Method:** Built a digital twin simulation to implement and test a Game Theory based protocol;
- **Outcome**: github.com/AdorableLake/Micro-V2X-Interaction

Data Science: Global E-waste Data Analysis & Visualization
Project Developer, Georgia Tech-CSE6242 **Jan. 2025 - Apr. 2025**

- **Proposal:** Analyzing and visualizing the growing gap between global e-waste generation and recycling to inform policy;
- **Method:** Led an end-to-end data science pipeline, from web scraping to developing an interactive network visualization using D3.js;
- **Outcome:** github.com/AdorableLake/Global-Ewaste-Analytics

Skills and Certifications

Programming: Python, C/C++, SQLite
Software: Tableau, Shapr3D, Axure, Sketch
Certifications: IBM Data Science (via Coursera), Apple Teacher
Language: English: TOEFL 93, BestScore of 97